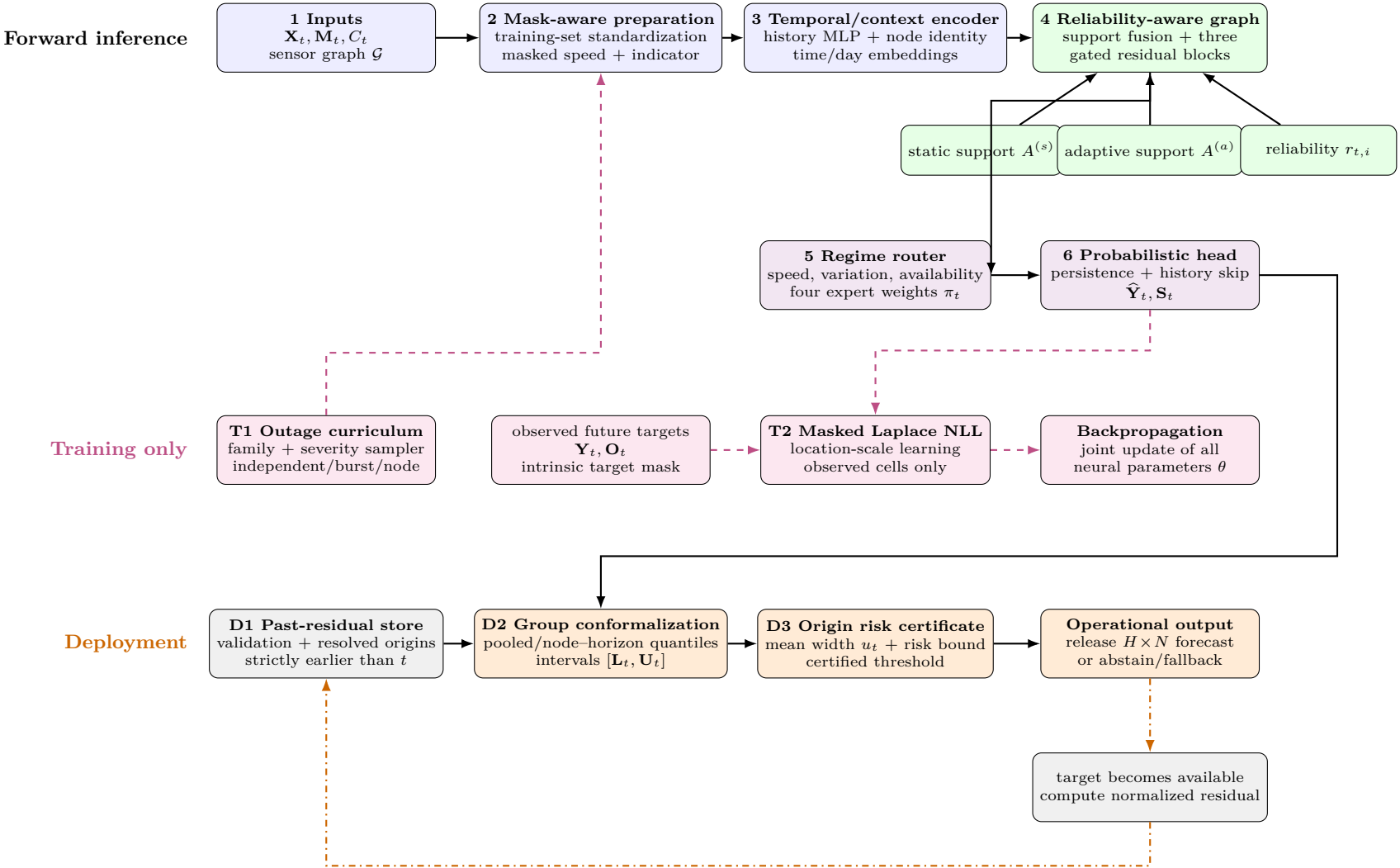
Forward inference
1 Inputs
$\mathbf{X}_t, \mathbf{M}_t, C_t$
sensor graph $\mathcal{G}$
2 Mask-aware preparation
training-set standardization
masked speed + indicator
3 Temporal/context encoder
history MLP + node identity
time/day embeddings
4 Reliability-aware graph
support fusion + three
gated residual blocks
static support $A^{(s)}$
adaptive support $A^{(a)}$
reliability $r_{t,i}$
5 Regime router
speed, variation, availability
four expert weights π_t
6 Probabilistic head
persistence + history skip
$\hat{\mathbf{Y}}_t, \mathbf{S}_t$
Training only
T1 Outage curriculum
family + severity sampler
independent/burst/node
observed future targets
$\mathbf{Y}_t, \mathbf{O}_t$
intrinsic target mask
T2 Masked Laplace NLL
location-scale learning
observed cells only
Backpropagation
joint update of all
neural parameters θ
Deployment
D1 Past-residual store
validation + resolved origins
strictly earlier than t
D2 Group conformalization
pooled/node–horizon quantiles
intervals $[\mathbf{L}_t, \mathbf{U}_t]$
D3 Origin risk certificate
mean width u_t + risk bound
certified threshold
Operational output
release $H \times N$ forecast
or abstain/fallback
target becomes available
compute normalized residual